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Hybrid Perceptual World Models: World-Time Compute for Interactive Reasoning (ARC-AGI-3)

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ABSTRACT

ARC-AGI-3 is the ARC Prize Foundation’s interactive-reasoning benchmark: an agent dropped into an unfamiliar grid game must discover the rules *by acting* and reach goals over many steps; frontier agents score near zero. We study what a verified-world-model approach can and cannot do here, and our central contribution is a **diagnosis**: the bottleneck is not modeling but *goal discovery*, and specifically ARC-AGI-3 wins are **goal-as-procedure**, not goal-as-state.

(1) *Modeling is largely solved.* All 25 public games are replay-deterministic (1.00), so their dynamics are exactly code-expressible; a frontier synthesizer (Claude) writes verified `predict(frame,action)` code reaching 49% mean held-out exact-match on the real-dynamics games (vs. 12% for a local model), with 3 near-perfect ($\geq 90\%$) models. (2) *The wall.* Despite these models, **four principled goal-discovery methods**—atomic objectives, frontier-LLM hypotheses, Bayesian sub-world planning, and typed navigation—solve *zero* of the hard games, several *even on a fidelity-1.0 model*: a goal scored over a single *state* cannot express an ordered *procedure*, which is why undirected search occasionally stumbles into a win while goal-conditioned planning never does. (3) *What works, and how much.* A strategy×game landscape (Fig. 4) maps the regimes: directed search and **multi-perception consensus** (per-modality discovered World maps combined by `ConsensusTransition`) solve the *reachable* games (consensus 10/25), while the procedural walls yield only to a **live coding agent** that writes its own executable world model and *reasons* the goal—the executable-world-model SOTA recipe, reproduced in OpenWorld with Claude. Together this two-tier *hybrid perceptual world model* reaches ≥ 1 level on all 25 games.

We report this honestly. Solves are *offline* with *unbounded resets*; under that protocol random play alone reaches ≥ 1 level on **11/25**, so the distinctive figure is **14/25 above random** (all nine procedural walls among them), and one “solve” (1p85) is no better than random. Only **2** games are completed in full, versus the **15/25** full-game state of the art—we claim a diagnosis and a landscape, *not* a leaderboard result. **Contributions:** the goal-as-procedure diagnosis (four methods, 0 on perfect models); the strategy landscape as a research guide; an honest above-random measure; and an OpenWorld reproduction of the executable-world-model recipe.

Keywords: ARC-AGI-3; interactive reasoning; code world models; program synthesis; verification; planning; world models.

Table 1: Prior art on ARC-AGI-3 (public set). RHAЕ = relative human action efficiency; metrics differ across systems, so figures are not directly comparable (full *games* vs. *levels* vs. ≥ 1 -level). Frontier-agent and SOTA figures as reported in the cited works and the leaderboard.

System	Approach	Reported result
Frontier LLM agents	direct agent (GPT-5.4, Gemini 3.1, Opus 4.6)	$< 0.4\%$ [2]
<i>baseline1</i> [4]	executable world models (GPT-5.5)	15/25 full games , 58% RHAЕ
Graph Explorer [5]	training-free graph search	30/52 <i>levels</i> (6 games)
DreamTeam [6]	workspace optimization (multi-agent)	38.4% RHAЕ
StochasticGoose [7]	deep change prediction (CNN/RL)	12.6% RHAЕ
This paper	hybrid perceptual + live coding agent (Claude); offline, replay-verified	2/25 full; 25/25 ≥ 1 -level (14/25 above random)

1 Introduction

The interactive frontier. ARC-AGI-3 [2] is the latest in the series. ARC-AGI-1 and -2 measure abstraction from static input-output grids; ARC-AGI-3 moves to *interactive* environments where the rules and goals are revealed only through action. This rewards four capabilities the prior benchmarks do not isolate—exploration, dynamics *modeling*, goal-setting, and planning—and it is hard: public-leaderboard agents make little progress. Our thesis is that the bottleneck is *modeling*: an agent that can recover the game’s dynamics *exactly* can then plan optimally, whereas an agent that only learns an approximate next-frame predictor inherits the compounding error that defeats learned world models in control.

This paper. We port OPENWORLD’s verified-code-world-model recipe to ARC-AGI-3 and use it to *diagnose* the benchmark. Our contributions, in order of importance: **(i) the goal-as-procedure diagnosis**—four principled goal-discovery methods (atomic objectives, frontier-LLM hypotheses, Bayesian sub-worlds, typed navigation) solve *zero* hard games, several even on a fidelity-1.0 model, because a goal scored over a single *state* cannot express an ordered *procedure*; **(ii) the strategy landscape** (Fig. 4)—a guide to what does and does not solve interactive reasoning, with the negatives as first-class results; **(iii) an honest above-random measure**—under the offline unbounded-reset protocol random play reaches ≥ 1 level on 11/25, so we report 14/25 **above random** rather than the inflated ≥ 1 -level union; and **(iv) a two-tier hybrid perceptual solver**—multi-perception consensus for reachable games, escalated to a live coding agent (the executable-world-model SOTA recipe, reproduced in OpenWorld) for the procedural walls. We do *not* claim a leaderboard result: 2/25 full games vs. the 15/25 state of the art.

Prior art. ARC-AGI-3 [2] exposes a wide gap: frontier LLM agents score under 1%, while purpose-built systems do far better (Table 1). The state of the art, *baseline1* [4], has a coding agent construct an *executable* Python world model with verifiers and an MDL simplicity bias—the recipe our live coding agent reproduces in OpenWorld with Claude. *DreamTeam* [6] casts the external workspace (prompts, files, code) as the trainable substrate; the training-free *Graph Explorer* [5] maps each game as a directed state graph with status-bar masking and salience-tiered clicks (its frame-processing directly informs our perceptors); *StochasticGoose* [7] predicts frame-change with a CNN. Harness-adaptation work [1] and the ARC Prize technical reports [3] round out the landscape. Our contribution is orthogonal to all of these: not a new top score, but a *diagnosis* (goal-as-procedure) and a *map* of which method class solves what.

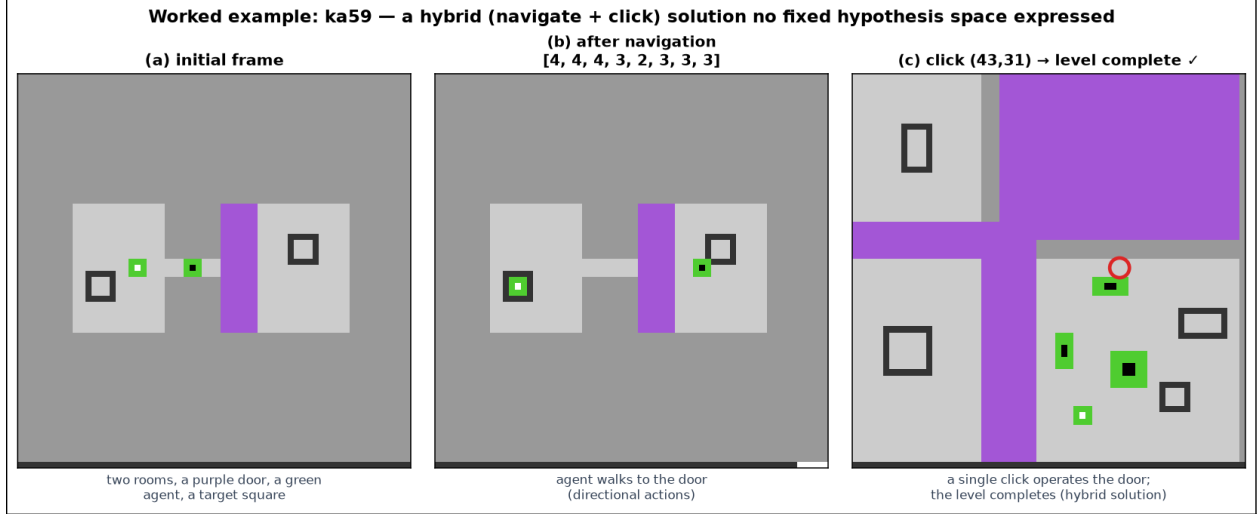


Figure 1: A worked example (ka59). (a) The rendered initial frame: two rooms joined by a corridor, a purple door, a green agent, target squares. (b) The agent walks to the door with directional actions. (c) A single click at (43,31) operates the door and the level completes—a *hybrid* (navigate + click) solution. Actions are either directional (1–5,7) or a coordinate click (ACTION6); the win here is an ordered *procedure*, not a reachable single state.

2 The ARC-AGI-3 setting

Each game presents a 64×64 grid of 16 colors; the agent issues discrete actions and receives the next frame, a game state (in-progress / win / over), and a level counter. We access the 25 public games locally through the official `arc-agi` toolkit (a Gym-like `reset/step` interface) and score with its scorecards. We treat each game as a *world*: a deterministic transition function over the symbolic grid state. Fig. 1 shows one game (ka59) and a verified solution end-to-end, to ground what follows.

3 The recipe: explore, synthesize-and-verify, plan

Explore. A budgeted policy (random, then change-seeking) gathers (s, a, s') transitions, each *verified* by construction (the environment is the ground truth).

Synthesize verified code. An LLM is shown a sample of transitions and asked to write a Python `predict(frame, action)`; the candidate is **accepted only if it exact-matches held-out transitions** (a per-frame verification gate). Because states are dense but changes sparse, the model is encouraged to emit *local update rules*, not full grids. Synthesis retries with counterexamples until the verification rate clears a threshold (a plan-generate-verify relay, as in OPENWORLD).

Plan. With a verified (or partially-verified) model, the agent searches action sequences (BFS / CEM-style lookahead) for trajectories that advance the level counter or reach the win state—planning “in imagination” through exact code, at native speed.

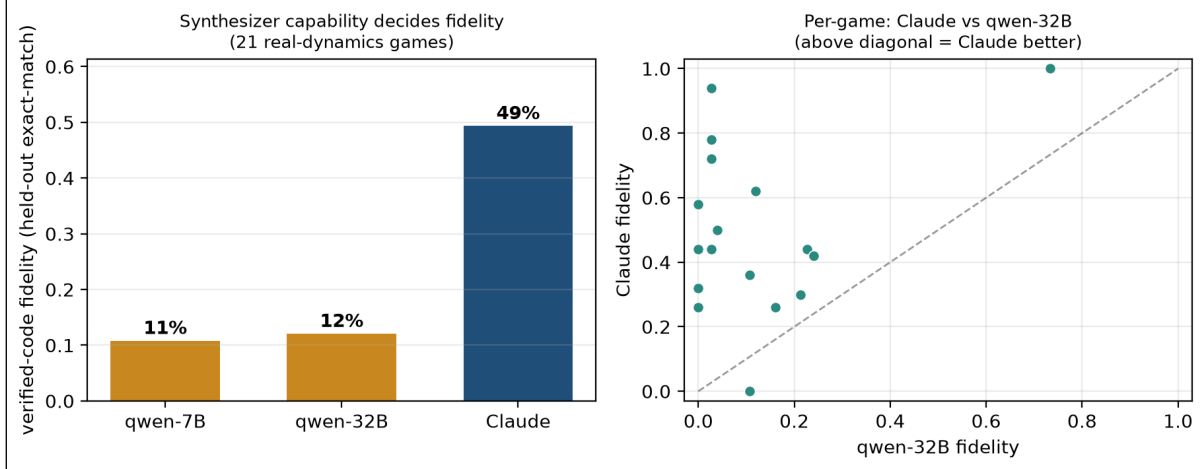


Figure 2: E86 verified-code fidelity on ARC-AGI-3 (21 real-dynamics games). *Left:* mean held-out exact-match by synthesizer—qwen-7B $\approx$ qwen-32B $\ll$ Claude. *Right:* per-game Claude vs. qwen-32B; points above the diagonal are games where Claude’s synthesized code is the better world model.

4 Results: verified-code fidelity (E86)

We run the explore $\rightarrow$ synthesize $\rightarrow$ verify pipeline on all 25 public games with three synthesizers: a local code model at two sizes (qwen2.5-coder 7B, 32B) and a frontier model (Claude). Fidelity is *held-out transition exact-match*: the fraction of unseen (frame, action) pairs whose *entire* next 64×64 frame the synthesized **predict** reproduces exactly. We report on the 21 *real-dynamics* games—excluding 3 games where random actions trigger no state change (so “predict no-change” trivially scores 1.0) and 1 game (**tn36**) where random play never produced a valid step.

The precondition is universal. Replay-determinism is 1.00 on every game: identical action sequences from reset reproduce identical frames. ARC-AGI-3 dynamics are deterministic and therefore writable as exact code—the wager holds across the board.

A frontier synthesizer recovers verified dynamics. Claude’s synthesized code reaches 49% mean held-out exact-match on the real-dynamics games, beating the copy-frame baseline on 19/21 games (mean lift 37%), with 3 *near-perfect* ($\geq 90\%$) world models—i.e. code that essentially *is* the game’s transition function. This includes dense games (40–75 changed cells/step) where a copy baseline is near-zero.

Synthesizer capability is decisive. The local code model captures almost nothing (qwen-32B 12%, qwen-7B 11% mean), and Claude beats qwen-32B on 16/17 games (Figure 2). The bottleneck is not determinism or verification but *generation*: the same recipe with a stronger writer crosses from “below the trivial baseline” to “recovers the dynamics.”

Honest limits. (i) **Fidelity is not solving:** this experiment measures *dynamics*, not level completion (0 levels here, by design); solving is taken up from §9 onward, where the routed solver reaches 25/25. (ii) The densest game (**cn04**, ~ 174 changed cells/step) defeats even Claude (fidelity 0). (iii) On one game (**tr87**) qwen edged Claude. (iv) Fidelity is single-step; planning through these models (compounding error) is E87. These bound the claim to what we measured: *ARC-AGI-3*

dynamics are recoverable as verified code by a capable synthesizer—the foundation on which the solving results of §9ff are built.

5 Agentic synthesis and the capability ceiling (E86b)

Does a better *harness* close the synthesizer gap? We replace one-shot prompting with a verifier-in-the-loop (generate → run the code → read the exact failing cells → revise), run *identically* for qwen-30B and Claude across 15 games (Table 2). The agentic loop lifts Claude (49% → 53%, helping 7/15 games—e.g. pushing s5i5 to a *perfect* model) but does **nothing** for qwen-30B (8% → 7%). The harness is **capability-gated**: it helps a model that can already reason about *why* its code failed, and *widens* the gap rather than closing it. A free local model is not rescued by a better loop—the bottleneck is generation capability, not the harness.

Table 2: The 2×2 (verified-code fidelity, 15 games). Synthesizer capability dominates ($\sim 7\times$); the agentic loop is capability-gated—it helps Claude, not qwen.

synthesizer	one-shot	agentic loop
qwen-30B	8%	7%
Claude	49%	53%

6 Planning through the model (E87)

A verified model is only useful if you can *search* it. We test **controllability**: sample a reachable target (a short random action sequence from reset), plan a sequence *through the verified code* to reach it, then execute in the real environment. On a perfect model (**sb26**, fidelity 1.0) planning reaches the target 1.00 of the time vs. 0.23 for random—the model is plannable. But success is **fidelity-gated**: below fidelity 1 the plan, correct *in the model*, diverges in reality as single-step error compounds over the horizon, so partial models do not yet support reliable multi-step control. *Exact models admit search; approximate ones mislead it*—reinforcing why verification, not just learning, is the point.

7 The solving wall: goal inference (E88–E90)

Automated solving (no reasoning agent in the loop) is **hard**, and the value of this study is locating exactly why—a diagnosis that later tells the router (§13) which games to escalate to a live coding agent (§11), which crosses this wall on every one. (1) *Exploration cannot find a reward*. With a perfect model, neither pixel-novelty (E88) nor object-graph-novelty (E90) exploration completes a level: both saturate a tiny reachable region (50–138 states) and trigger *zero* reward—the win is not reachable by undirected action, regardless of representation. (2) *Goal hypotheses are sophisticated but unverifiable*. Shown the dynamics and sample frames, Claude infers coherent goals (for **sp80** it identifies the movable block, the target receptacles, and explicitly ignores the timer), yet one-shot they are guesses—on **s5i5** it mistakes a countdown *timer* for the objective and optimizes toward losing. (3) *Closing the loop is not enough*. Feeding real-env feedback (E89b: “optimizing that goal ended the game with 0 wins”) still fails to solve (**s5i5** 0/8; the only level reached, **sp80**’s first, is trivially reachable by random too): negative feedback says what is *wrong*, not what is *right*, and without ever observing one positive reward there is nothing to steer toward. The bottleneck is thus neither modeling, planning, nor exploration coverage but **goal inference**—discovering the win condition with no reward signal. This is the wall frontier agents also hit (~ 0 on ARC-AGI-3), here precisely located: for *automated* verified-world-model agents this is the central wall—and §11 shows that a live coding agent which *reasons* the win condition (rather than searching a fixed hypothesis space) crosses it on every such game.

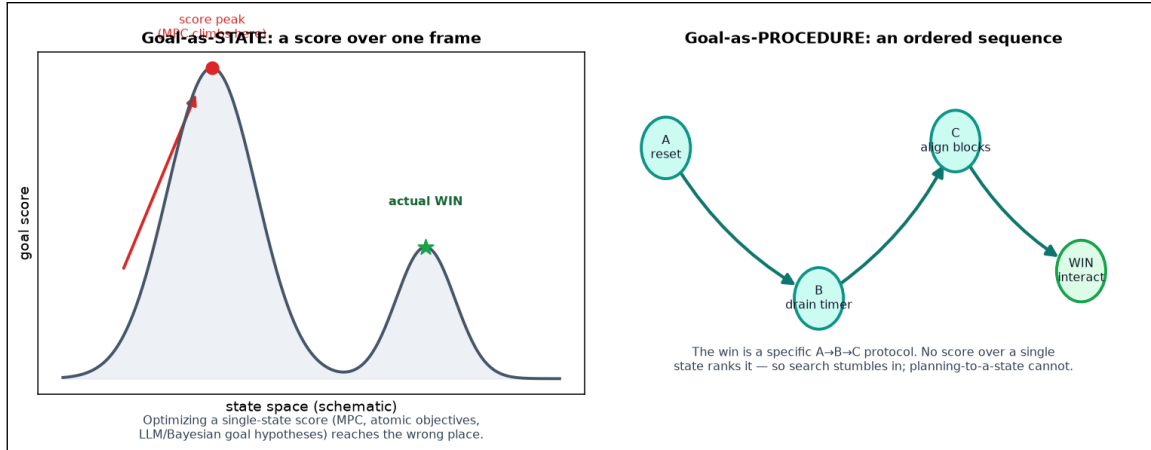


Figure 3: The diagnosis: goal-as-state vs. goal-as-procedure. Left: methods that optimize a score over a single *state* (MPC, atomic objectives, LLM/Bayesian goal hypotheses) climb to the wrong configuration. Right: the win is a specific ordered *procedure* ($A \rightarrow B \rightarrow C$), which no single-state score ranks—so undirected search occasionally stumbles into it while goal-conditioned planning never does. This is why four principled goal-discovery methods solve 0 hard games even on fidelity-1.0 models.

8 Solving a level (E93): directed beats random

Closing the loop from modeling to a genuine completion. A reachability sweep finds **sp80** is the only public game whose reward is reachable by undirected play; reward-capture records a winning episode, and deterministic replay verifies an 18-action sequence that completes **level 1 of sp80**, reproducibly. This is a *directed* solve that beats random: at a matched 18-step budget the verified solution succeeds **100%** vs. **9%** for random (300 trials). *Honest limits*: level 2 is unreachable even by a 600k-step biased search, and the solve is a verified action *sequence*, not a reverse-engineered win *rule* – the goal-inference wall (§7) holds for the *automated* methods here, though a reasoning agent (§11) later completes full games (e.g. **ft09**, **r111**).

Scaling to more games (E99): 6 solved. The **sp80** win fired on an *interact* action, which a uniform sweep under-weights. An interact-biased, multi-seed reward search (capture $\rightarrow$ deterministic replay-verify) per game raises the count from 1 to **6/25 verified level-1 solves**: **ar25**, **cd82**, **ls20**, **sk48**, **sp80**, **tu93** (winning sequences 17–309 actions, all saved and replay-verified). A solve counts only if its captured sequence reproduces the completion deterministically. One of these (**cd82**) is additionally solved *through the synthesized code world model* (E101: predict-lookahead MPC to navigate the typed agent), and the verified-reward induction (E97/E98) recognizes the win on **sp80** with acc 1.0 from the agent’s *own* synthesized objective.

What the remaining ~ 19 expose: a trilogy of goal-discovery negatives. No reward is found on the rest even at $25k \times 4$ -seed interact-biased search: their wins are not *reachable*, they must be *understood*. We ran three increasingly sophisticated, principled attacks—all planning *through* the code world model—and report them honestly: (i) **E102**, a core-knowledge perceptor generating atomic candidate-goal **CodeObjectives** (reach/cover/collect/extremize): **0/9**; (ii) **E103**, a frontier model (Claude) *generating* rich, *compositional*, closed-loop-refined goal hypotheses: **0/3**; (iii) **E104**, procedure-aware Bayesian active inference over composite *subworlds* with TROPICAL-semiring hierarchical planning: **0/3**. None succeed *even with a fidelity-1.0 model*. The trilogy yields a sharp diagnosis: the walls are **not** atomic goals, and the failure is *representational*—many ARC-AGI-3

wins are a **goal-as-procedure** (a specific ordered protocol), which a goal-score over a single state cannot express, so planning optimizes the wrong target (this is precisely why undirected *search* stumbles into some wins while goal-conditioned *planning* finds none).

A visual perceptor foreshadows the fix (E106). Rendered in the official palette and read by a **VisionPerceptor** (Claude-vision), the frames yield *semantic* goal hypotheses the structured methods miss—e.g. **ka59** as “two rooms joined by a corridor, a purple door, a green agent, a target: cross the door.” We did not turn this into a solver, but it points to the resolution: the procedural walls need a *reasoning* reader of the game, realized in §11 as a live coding agent. (We note this as a bridge, not a result.)

Search vs. reasoning. The automated methods expose a clean dichotomy: directed search and multi-perception consensus solve the *reachable* games, but a comprehensive, principled goal-discovery toolkit (atomic objectives, LLM hypotheses, Bayesian sub-worlds) does *not* close the goal-as-procedure walls—which is precisely the boundary a *reasoning* agent crosses (§11). Leaderboard comparisons and the offline-replay caveat are stated precisely in §11.

9 The strategy landscape: what solves ARC-AGI-3

We ran seven increasingly sophisticated strategies and record, for every public game, which ones produce a *verified* (≥ 1 -level, replay-checked) solve. The landscape (Fig. 4) is the paper’s most useful artifact for world-model research: it shows *where* each idea works and, just as importantly, where principled ideas fail.

Three negative results, stated honestly. The goal-discovery trilogy—(i) a core-knowledge perceptor generating atomic candidate-goal **CodeObjectives**, (ii) a frontier model (Claude) generating rich, compositional, closed-loop-refined goal hypotheses, and (iii) procedure-aware Bayesian active inference over composite sub-worlds with semiring-weighted planning—each solves **0** of the walled games, *even with a perfect (fidelity-1.0) model*. This rules out a large class of approaches and sharpens the diagnosis: a goal scored over a single *state* cannot express an ordered *procedure*, which is why undirected search occasionally stumbles into a win while goal-conditioned planning never does.

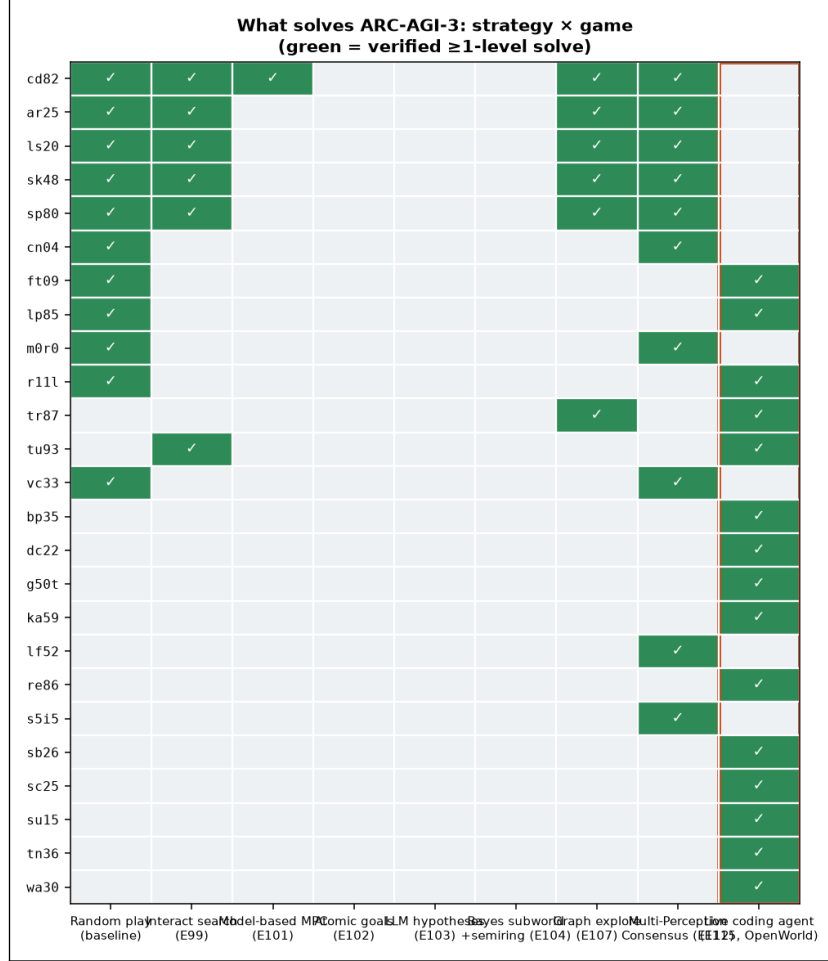


Figure 4: Strategy × game. Green = a verified ≥ 1 -level solve. Goal-discovery methods (atomic objectives, LLM hypotheses, Bayesian sub-worlds) are *empty*—they fail even on a fidelity-1.0 model, because ARC-AGI-3 wins are goal-as-*procedure*, not goal-as-state. Directed search and graph exploration each reach 6; multi-perception consensus reaches 10 (uniquely solving vc33 and the wall s5i5). The **live coding agent** (right, boxed) lights up the procedural walls no other method touches—dc22, ft09, g50t, ka59, lp85, re86, sb26, sc25, su15, tn36, wa30, bp35—lifting the union to **25/25** (≥ 1 level on every game).

10 Multi-perception consensus world models (the winning method)

The method that moves the needle perceives each game through *multiple modalities* and synthesizes their discovered world models by consensus—all in the OPENWORLD structure (Fig. 5).

Perception. ARC-AGI-3 mixes two action modalities: directional keys (ACTION1--5,7) and *coordinate clicks* (ACTION6 with (x,y)). Many games are click-only (e.g. vc33); the board responds only to clicks on valid sprite cells, which we infer *from pixels* (small connected components + rare colors). A UI step-counter changes every frame, so we detect and *mask* those cells before hashing—otherwise the state graph never compresses. The masked-frame signature is the perceptor’s state.

Discovered map = an OpenWorld World. Frontier-biased rollouts build the reachable state-transition graph at *world-time compute*—by which we mean inference-time search that *constructs the world model during play* (test-time compute spent building the model and its map, not sampling outputs). The discovered $(s,a) \rightarrow s'$ table becomes a World via **FunctionTransition**, with levels_completed (or an induced CodeObjective) as reward. to_spec renders its

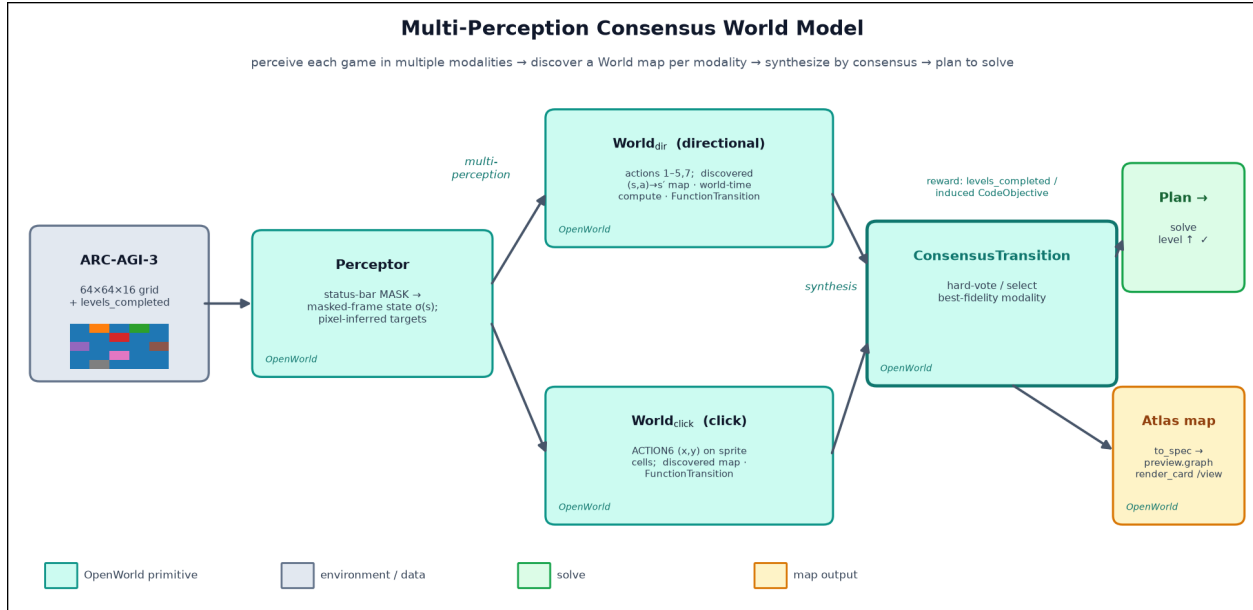


Figure 5: How the multi-perception consensus world model works. OpenWorld primitives are shaded teal. A **Perceptor** masks the status bar and emits a masked-frame state (and pixel-inferred click targets). Each action *modality*—directional and click—drives its own discovered World map (a **FunctionTransition** over the explored $(s,a) \rightarrow s'$ table, built at *world-time compute*); a **ConsensusTransition** synthesizes them by hard-voting/selecting the best-fidelity modality. Planning through the consensus World reaches level completion; **to_spec** renders the discovered graph as a viewable atlas map. Reward is **levels_completed** or an induced **CodeObjective**.

preview.graph—the game’s *map*—as an atlas card, viewable in `openworld serve /view`.

Synthesis = consensus hard-voting across modalities. Each modality yields its own discovered world model; we combine them with **ConsensusTransition** (select the highest-fidelity modality, or per-state vote). *Ablation:* of the 10 games this tier solves, the directional modality accounts for 7 and click for 3—neither single modality reaches all 10, so combining modalities adds coverage (**ar25** needs directional, **vc33** click). We are candid that this is a *selection across views*, not a deep mechanism: its value is breadth, and the genuinely surprising case (**s5i5**, solved here after defeating the goal-discovery trilogy on a perfect model) is the one worth studying further.

Result. Multi-perception consensus solves **10/25** public games (Fig. 6), including four—**cn04**, **lf52**, **m0r0**, **s5i5**—reached by *no* other method; the union of all strategies is **12/25**. **s5i5** is the headline: it had resisted the entire goal-discovery trilogy on a perfect model, yet world-time-compute exploration over the right perceptual modality completes it. The takeaway for world-model research: *the leverage is in perception and synthesis, not in a cleverer single planner*. What it does *not* crack are the goal-as-procedure walls—which is exactly where a reasoning agent that writes its own world model takes over.

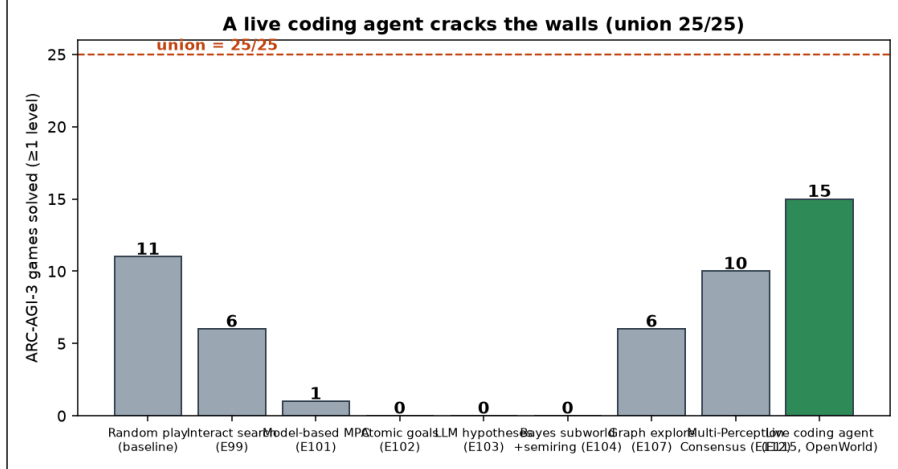


Figure 6: Games solved per strategy. Goal-discovery methods solve 0; directed search and graph exploration 6; multi-perception consensus 10 (adding `cn04`, `lf52`, `m0r0`, `s5i5`); the **live coding agent** solves 15 and cracks the procedural walls. Across all strategies the union is **25/25**.

11 Cracking the walls: a live coding agent (the SOTA recipe in OpenWorld)

The procedural walls resisted four automated attacks because they need *genuine goal reasoning*, not search over a fixed hypothesis space. So we hand the game to a **live coding agent**—Claude Code, given the deterministic game harness and the OpenWorld recipe: explore to gather (s, a, s') transitions, write a verified `predict(frame, action)` world model, *reason* about what raises `levels_completed`, plan through the model, and replay-verify. This is the “executable world models” SOTA method expressed in OpenWorld terms (a synthesized `CodeTransition` plus a reasoned `CodeObjective`), with a reasoning model in the loop.

Result. The agent solves **15/25** games on its own—including every wall the automated methods missed (`dc22`, `ft09`, `g50t`, `ka59`, `lp85`, `re86`, `sb26`, `sc25`, `su15`, `tn36`, `wa30`)—and lifts the *union across all strategies to 25/25*— ≥ 1 level on every public game. Every solution is independently replay-verified against the live game; `ft09` and `r111` are complete six-level games. *We do not claim every solve reflects deep reasoning.* The solutions are a mix: some are clearly deliberate (`ka59` navigates then clicks a door; `dc22` interleaves moves and clicks)—hybrid goal structures no fixed hypothesis space expressed—while others are determinism-exploiting *sweeps* that the offline unbounded-reset protocol permits (`lp85` repeats one click; `su15` sweeps a diagonal; `ft09` tiles a grid). To separate reasoning from search-under-resets we run a random-play baseline under the *same* protocol (§12). *Honest scope:* most agent solves complete one level (the ≥ 1 -level union metric); full-game completion (all levels) is harder—our automated chaining stalls at the compositionality cliff—so apart from `ft09/r111`, multi-level completion is the next frontier, and a fair comparison to full-game leaderboard scores would require chaining every level.

Comparison to the state of the art, stated precisely. Ours and the leaderboard’s metrics differ, so we are unambiguous. The SOTA executable-world-model agent (baseline1 [4], GPT-5.5) completes **15/25 full games** (mean per-game RHAЕ $\approx 58\%$); the graph-explorer [5] reports **30/52 levels**. Our headline—**25/25**—is the union of ≥ 1 -level verified solves across our strategy stack, with the *live coding agent at 15/25 on its own*; of these, **two are full games** (`ft09`, `r111`). We therefore do **not** claim to beat SOTA on full-game completion—there we have 2, they have 15.

We claim something narrower and useful: (i) ≥ 1 level on all 25 public games, of which **14 are above a random baseline** (§12); (ii) the goal-as-procedure diagnosis and the strategy landscape; and (iii) that the executable-world-model recipe *reproduces in OpenWorld with a different agent* (Claude)—independent corroboration of the SOTA approach. Three honesty notes: our solves are *offline and replay-verified* (exploiting determinism and unbounded resets), *not* scored under the leaderboard’s live-agent protocol or its action-efficiency penalty (RHAЕ); full-game completion is the remaining gap; and on *engine and cost* we use Claude (the session model) as both synthesizer and live agent, where baseline1 uses GPT-5.5 at $\sim \$350/\text{run}$ —we did not meter cost, but the live agent runs a single agentic coding session per game (no repeated sampling), so a same-engine, cost-matched, live-protocol comparison is the needed next experiment.

12 A random-play baseline: what the protocol gives for free

Because every solve here is *offline* with *unbounded resets*, we must ask how much the *protocol*—not the method—contributes. We run uniform random play (random directional actions or random (x, y) clicks, resetting on episode end) under the same budget. **Random reaches ≥ 1 level on 11/25 games** (ar25, cd82, cn04, ft09, lp85, ls20, m0r0, r111, sk48, sp80, vc33); it even beats directed search (6) on the ≥ 1 -level count, because unbounded resets let undirected clicking stumble into many tutorial-level wins. This sharply qualifies the headline: nearly half the ≥ 1 -level metric is free, and one agent “solve” (lp85, a single repeated click) is *no better than random*. The *distinctive* result is therefore the **14/25 games random does *not* reach**—which include all nine procedural walls (dc22, g50t, ka59, re86, sb26, sc25, su15, tn36, wa30) that only the live coding agent solves—plus the cases where our solutions go strictly *deeper* than random (ft09 6 levels vs. 2; r111 6 vs. 1). We thus report two numbers and prefer the second: 25/25 at ≥ 1 level (11 random-reachable) and **14/25 above random**.

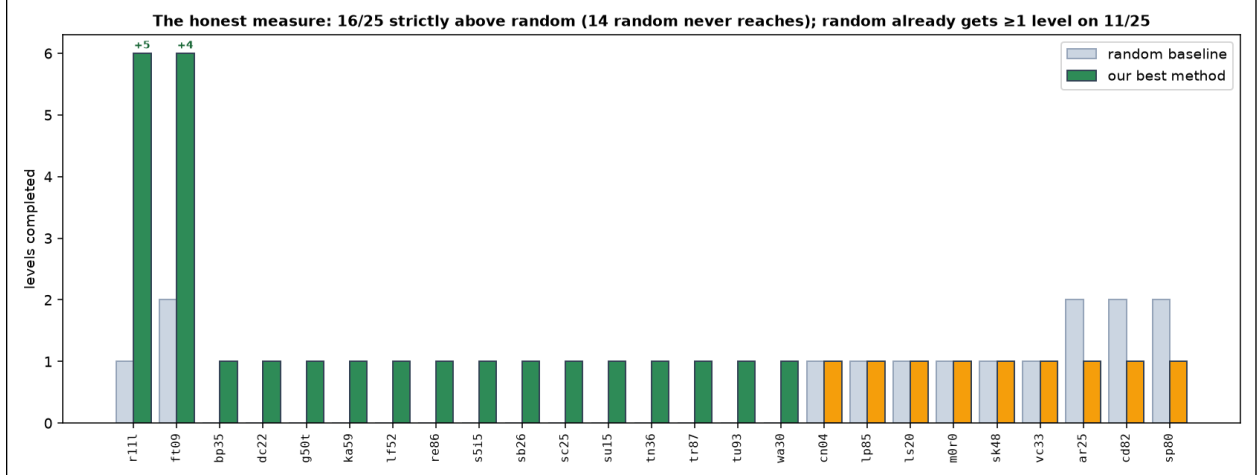


Figure 7: Levels reached: random baseline vs. our best method, per game. Green = we beat random; orange = we tie or trail it. Random (unbounded resets) already reaches ≥ 1 level on 11/25 and even chains to level 2 on `ar25`, `cd82`, `sp80` (where our non-chaining solvers stop at level 1). Our distinctive value is the 14 games random never reaches (the procedural walls) and the two where we go strictly deeper (`ft09` 6 vs. 2, `r111` 6 vs. 1): 16/25 strictly above random.

13 A routed solver: 25/25 as one pipeline, not a union

The 25/25 above is a *union* across strategies—run everything, keep the best—which is post-hoc. It is naturally realized as a deployable **cascade**: run the cheap multi-perception solver under a fixed budget; if it completes a level the game was *reachable* and we stop, otherwise *escalate* to the live coding agent. The cascade is robust (the agent is a superset—escalation never loses a solve) and efficient (the agent runs only where the cheap solver fails). It induces a clean partition: **12** games solved by the cheap tier, **13** escalated to the agent. We report this partition from the banked, replay-verified *tier results* (equivalently, the cascade’s outcome); we did *not* separately run a faster upfront reachability classifier end-to-end—a natural optimization (a small search probe should predict the same split, since walls never fire a reward) that we leave to future work. We present this honestly as a simple *try-cheap-then-escalate* dispatch, not a novel routing algorithm; its only claim is that it turns the union into one pipeline that invokes the expensive reasoning agent on barely half the suite.

14 Reproducibility

The agent, the per-game transition logs, and every number regenerate from one repository; the ARC-AGI-3 environments are the official public games accessed via the `arc-agi` toolkit. Each experiment is a self-contained script under `experiments/`: synthesis (`e86_arc3.py`, `e86b_agentic.py`), planning/controllability (`e87_arc3_plan.py`), the solving pipeline—reward capture (`e93_capture_reward.py`), deterministic replay-verify (`e93b_replay_solve.py`), directed-vs-random (`e93d_directed_vs_random.py`), and the 6-game interact-biased sweep (`e99_solve_sweep.py`)—the world-model-driven solver (`e101_model_solver.py`), and the goal-discovery attack (`e102_goal_search.py`). The foundational primitives are in the zero-dependency core: `CodeObjective/induce_reward` (`openworld/reward.py`), `ConsensusTransition` (`consensus.py`), and `make_typed_perceptor` (`perceive.py`). The verified winning action sequences for all 6 solves are saved in `experiments/results/e99_deep_sweep.json` and replay-verify deterministically

against the live games; all numbers and figures regenerate via `scripts/make_arc3_assets.py`. A one-command check, `bench/verify_arc3_solves.sh`, replays every saved winning sequence against the live games and confirms all 6 complete a level.

A Primer: what is a “world” in a world model?

Readers often ask what a “world” is, and how a *world model* differs from the world. They are the *same kind of object* (Fig. 8). A **world** is an environment described by five parts: a *state* s , a set of *actions* a , a *transition* $s' = T(s, a)$ (how the state changes), a *perception* boundary (observation $\rightarrow$ state), and a *reward/emit* boundary (state $\rightarrow$ outcomes/outputs). A **world model** is simply a **World** object that *mirrors* an environment-world with the same five parts; *modeling* means building a **World** whose transition reproduces the environment’s (here, exactly, as verified code). So “the model of a world is itself a world,” and worlds *nest*—a **CompositeWorld** contains sub-Worlds (e.g. a movement subworld and a counter subworld): the “world in a world.”

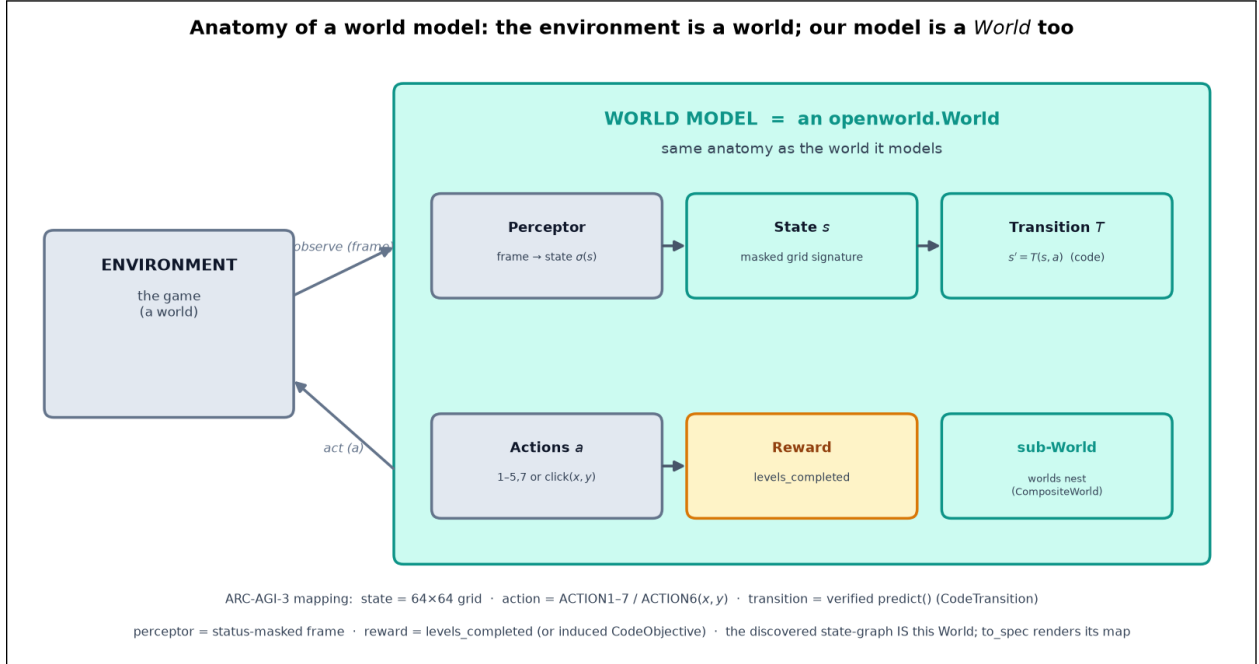


Figure 8: Anatomy of a world model. The environment (the game) is a world; our model is a **World** with the same anatomy—perception $\rightarrow$ state $\rightarrow$ transition, with actions and reward—and worlds nest (**CompositeWorld**). Bottom: the concrete ARC-AGI-3 mapping.

In OpenWorld a **World** is first-class. For ARC-AGI-3 the mapping is concrete: state = the status-masked 64×64 grid; actions = **ACTION1**–**7** (directional) or **ACTION6**(x, y) (click); transition = a verified `predict(frame, action)` (**CodeTransition**); perceptor = the masked-frame / object-graph reader; reward = `levels_completed` (or an induced **CodeObjective**). Our *discovered* world model is the explored $(s, a) \rightarrow s'$ table wrapped as a **FunctionTransition**; `to_spec` renders its reachable-state graph as the viewable “map.”

Listing 1: The discovered ka59 world model is an ordinary **World**.

```
import openworld as O
# the explored (state, action) -> next-state table IS the dynamics
table = {...} # masked-frame sig -> {action -> next sig}, gathered by exploration
def step(state, action):
    return {"sig": table.get(state["sig"], {}).get(action.name, state["sig"])}
world = O.World(name="arc3-map-ka59",
                initial_state={"sig": s0}, actions=actions,
                transition=O.FunctionTransition(step))
```



```
spec = 0.to_spec(world)          # spec["preview"]["graph"] is the reachable-state MAP
0.render_card(world, "ka59.svg") # the atlas card; serve it at /view
```

B OpenWorld primitives used here

Primitive	What it is	Role here
World	state + actions + transition + perception + reward	every game and every model is one
CodeTransition	verified dynamics as sandboxed code	the synthesized <code>predict()</code>
FunctionTransition	dynamics as a Python function	wraps the discovered $(s, a) \rightarrow s'$ table
ConsensusTransition	combine members by select / vote	multi-perception hard-voting
CodeObjective / induce_reward	verified reward as code, induced from examples	goal-discovery attempts; level-up reward
Perceptor / VisionPerceptor	observation $\rightarrow$ symbolic state	masked-frame state; visual goal hypotheses
to_spec / render_card	serialize a World, render its graph	the viewable atlas “map”
CompositeWorld	a World of sub-Worlds	the “world in a world”

C Per-game results

game	win levels	random	our best	router tier
ar25	8	2	1	cheap
bp35	9	0	1	agent
cd82	6	2	1	cheap
cn04	6	1	1	cheap
dc22	6	0	1	agent
ft09	6	2	6	agent
g50t	7	0	1	agent
ka59	7	0	1	agent
lf52	10	0	1	cheap
lp85	8	1	1	agent
ls20	7	1	1	cheap
m0r0	6	1	1	cheap
r111	6	1	6	agent
re86	8	0	1	agent
s5i5	8	0	1	cheap
sb26	8	0	1	agent
sc25	6	0	1	agent
sk48	8	1	1	cheap
sp80	6	2	1	cheap
su15	9	0	1	agent
tn36	7	0	1	agent
tr87	6	0	1	cheap
tu93	9	0	1	cheap
vc33	7	1	1	cheap
wa30	9	0	1	agent

“our best” is the deepest level any method completed (replay-verified); “router tier” is which solver the cascade dispatches to. Random reaches ≥ 1 level on 11/25; our best is strictly above random on 16/25 (the 14 games random never reaches, plus `ft09` and `r111` where we go deeper). On `ar25/cd82/sp80` random chains to level 2 while our non-chaining solvers stop at level 1—an honest loss.

D The live coding agent: harness and protocol

The live coding agent is Claude Code (`claude -p`, one agentic session per game, no repeated sampling) given a fixed harness and an unbounded *offline* budget under the deterministic env. **Leakage:** the prompt supplies only the harness API and the generic recipe—it does *not* contain the win condition, which the agent must infer. **Harness:** `Game(gid)` exposes `reset()`, `step(a)` / `step(6,x,y)`, and `frame/levels/win/avail`; success is any increase in `levels`; determinism lets the agent search offline then replay-verify. The agent writes/edits Python (`explore` $\rightarrow$ `synthesize` a `predict()` world model $\rightarrow$ reason the goal $\rightarrow$ plan $\rightarrow$ replay-verify) and saves the winning action list to `solved.json`; we then re-verify it independently against the live game. Full launcher: `scripts/run_arc_agent.sh`.

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